

Gene Pulser® Electroporation

Cell Type	Bacterial, gram positive	Molecules Electroporated	DNA: pGT633, covalently closed circular form, a native 9.8kb erythromycin resistant <i>Lactobacillus</i> plasmid.
Species Used	<i>Lactobacillus reuteri</i> , 100-63(3) a gastrointestinal isolate from fowl crop		

Before the Pulse

Cell Growth Medium	Lactobacilli MRS broth (Difco)	Growth Phase at Harvest	O.D. (600) = 0.8 (log phase cells)
		Pre-pulse Incubation	0°C for 1 min

Wash Solution 3.5X SMEB; Luchansky *et.al.* (1988) Bio-Rad Bulletin **1350**:1-3

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	0 °C		
Electroporation Medium	3.5 x SMEB (1x = 272 mM sucrose, 1 mM EDTA)	Cuvette Gap	0.4 cm
Cell Density	10 (9) cells / ml	Voltage	2.5 kV
Volume of Cells	800 µl	Field Strength	6.25 kV/cm
DNA Concentration	10 µg		
DNA Resuspension Buffer	TE (10 mM Tris, 1 mM EDTA, pH 8.0)	Capacitor	25 µF
Volume of DNA	5 µl	Resistor	(Pulse Controller) Not used**
After the Pulse		Time Constant	10 to 15 msec
Outgrowth Medium	Lactobacilli MRS broth (Difco) 10 ml		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.
Ref: H.J. Connell, "Investigation of Methods for the Transformation of Gastrointestinal Strains of *Lactobacilli* with Plasmid pGT633" (1990). Ph.D. thesis, University of Otago, Dunedin, NEW ZEALAND (2) This work was carried out under the supervision of Dr. G. Tannock, Dept. of Micro., University of Otago, Box 56, Dunedin, NEW ZEALAND. PH: +64-3-4797713, Fax 64-3- 4741607. Questions regarding the availability of strains and the pGT633 should be directed to him. **It is NOT RECOMMENDED to use high voltage with out the Pulse Controller.

Outgrowth Temperature	37 °C
Length of Incubation	3 hours
Selection Method or Assay Used	Erythromycin 25 µg / ml; requires a minimum expression time of 3 hr. to recover transformants
Electroporation Efficiency	1.6 x 10 (1) transformant / µg DNA (avg); 1x10 (1) to 1.2x10 (4) range
Per Cent Survival	17%

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071